

Yizheng Xie

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Education

Brown University , Providence, RI, USA Ph.D. in Computer Science (Distributed System, Operating System, Cloud Computing)	Sept 2023 – Expected May 2028
Boston University , Boston, MA, USA M.S. in Computer Science	Sept 2021 – May 2023
Chinese University of Hong Kong, Shenzhen , Shenzhen, Guangdong, China B.E. in Electronic Information Engineering	Sept 2017 – May 2021

Working Experience

Meta Platforms , Menlo Park, CA <i>Production Engineer Intern in Kernel Team (Python, Fedora Linux, Upstream Rust Packaging, Linux Userspace)</i>	May 2022 – Aug 2022
- Integrate Fedora and CentOS Stream forges and issue trackers (Pagure, GitLab, Bugzilla) into Meta Internal Task tool. - Collaborate with Fedora engineers to fix and improve external tools like fixing Bugzilla authentication issues and adding Copr build integration in fedora-create-review tool. - Implement a CLI tool to automate the process of updating Rust RPM packages by recursively updating dependencies and packaging missing dependencies, analyzing parallel chain-build order and checking compatibility for the current update set or any Bodhi rust update set. (Pagure Repository: pagure.io/fedora-rust/rust-update-set) - Integration between Meta and Upstream aims to help Meta employees contribute to Fedora and Stream, rust packaging tool aims to increase efficiency and safety for updating a large set of Rust packages for the whole Fedora community.	

Research

Slowpoke: End-to-end Throughput Optimization Modeling for Microservices <i>To appear NSDI'26, Project Lead (Distributed profiling, Tracing, Kubernetes, Istio, Docker, Golang, C++)</i>	Sept 2023 – Apr 2025
- Introduce the first profiling system for accurate what-if analysis of throughput optimizations in complex microservices, e.g., "what end-to-end throughput could be if I optimize the cart service from 10K req/s to 20K req/s?", achieving RMSE of 2.1% for 4 real benchmarks and 1.9% for 108 synthetic benchmarks covering comprehensive characteristics. - Design a performance model to infer bottleneck equivalence by selectively slowing down non-target services.	
λEASH: Scaling Out Shell Scripts with Recoverable Serverless Computing <i>Under submission of EuroSys'26, Project Lead (Serverless, Shell, AWS Lambda, Python, C, Rust)</i>	Sept 2023 – Jan 2025
- Introduce a system automatically scale out unmodified shell scripts correctly, achieving significant speedups over Bash (avg: 15.2×, max: 215.8×) with minimal cost increases (avg: \$0.11)—in some cases being both faster and cheaper. - Design a recoverability protocol to allow Lambda run longer than 15 minutes—limit imposed by AWS—while preserving correctness and Unix pipe-like streaming efficiency with minimal overhead (avg: 0.03× slower). - Schedule computations smartly across EC2 and Lambda to overcome limited network bandwidth, allow direct TCP communication between Lambdas using hole-punching technique, and automatically resolve software dependencies.	
Per-key Watermark for Apache Flink <i>Project Lead (Apache Flink, Java, Stream Processing)</i>	Sept 2022 – Jan 2023
- Modify Apache Flink source codes to support Per-key Watermark propagation, processing and API, generate testing sources with different number of keys, portion of delayed keys and delayed time interval. - Reduce latency of stream processing with 10-second tumbling window by 30% given 40000 keys with 50% delayed by 50ms, and the latency decreases more as the number of keys, portion of delays and delayed time increase.	

Publications

- [1] Y. Xie, D. Jin, O. Çölkesen, V. Kalavri, J. Liagouris, and N. Vasilakis, "Slowpoke: End-to-end throughput optimization modeling for microservice applications," in *23rd USENIX Symposium on Networked Systems Design and Implementation*, (Renton, WA), USENIX Association, May 2026.
- [2] Z. Huang, R. Dunder, Y. Xie, K. Kallas, and N. Vasilakis, "Fractal: Fault-tolerant shell-script distribution," in *23rd USENIX Symposium on Networked Systems Design and Implementation*, (Renton, WA), USENIX Association, May 2026.
- [3] Y. Xie, D. Jin, and N. Vasilakis, "Towards hybrid cooperative-preemptive scheduling," in *Proceedings of the 13th Workshop on Programming Languages and Operating Systems*, 2025.